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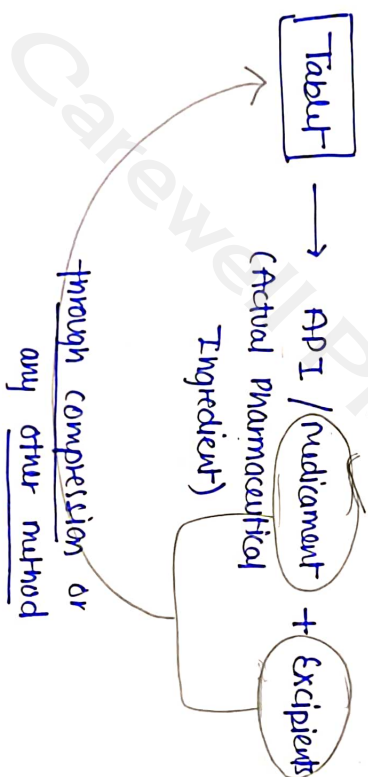
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TABLETS

most imp. in all industrial pharmacy

Tablets are compressed solid unit dosage form which contain medicament with or without excipients.



- Tablets are present in a different-2 shapes and sizes-- such as circular, flat, biconvex etc.



• Advantage of Tablets

- Simplest oral dosage form
- Easy to administered
- Cheap (not expensive).
- greatest dose precision.
- Unpleasant and bitter taste can be masked by sugar coating
- light in weight (easy to transport)
- Physically and chemically stable
- Sustained release product is possible by enteric coating

• Disadvantage of Tablets

- Infant, childrens and elder person can not easily administered
- Not useful for unconscious patients.
- Not used in emergency (slow action).
- Problem with compression to crystalline drugs.

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• Ideal characteristics of tablets :-

- A pharmaceutical tablet should have elegant product, which have proper shape and size (appearance).
- Tablet should be free from any defects like chips, cracks, discoloration and contamination.
- Tablet should have sufficient strength to withstand mechanical shock during its production, packaging, shipping and dispensing.
- Tablets must be uniform in weight and in drug content.
- Tablet must be able to release the medicament in a predictable and reproducible manner and get absorb into systemic circulation.
- Tablet and its medicament/exipient should have chemically and physically stable over-time.

• Classification of Tablets :-

1) Tablet ingested orally :

- i) compression tablet eg. Paracetamol tablet
- ii) Multiple compressed tablet eg. Coldwin
- iii) Sustained tablet
- iv) Enteric coated tablet (ECT)
- v) Sugar coated tablet
- vi) Film coated tablet
- vii) Chewable tablet

2) Tablet used in oral cavity :

- i) Buccal tablet
- ii) Sublingual tablet
- iii) Trochus or lozenge
- iv) Dental cone

3) Tablet administered by other route :

- i) Implantation tablet
- ii) Vaginal tablet

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4) Tablet used to prepare solution :

- i) Effervescent tablet
- ii) Dispensing tablet
- iii) Hypodermic tablet

1) Tablet ingested orally :-

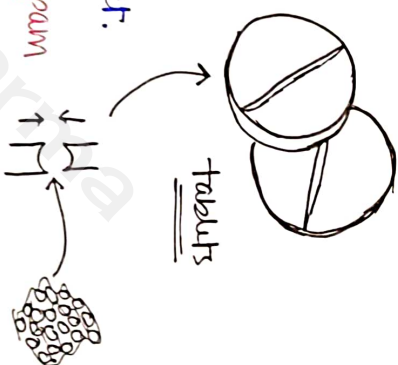
These are those tablets which are introduced in our body through oral route -

i) Compression tablets →

These tablets which are generally prepared by compression of powdered, granulation material with the help of punching machine.

- No special coating
- mostly water soluble.
- Raped disintegration
- and Release of product.

eg Paracetamol, diazepam



ii) Multiple compressed tablet →

- Those tablets which are composed of two or more different layer.
- used for those tablet which has two or more incompatible substance with each other.
- Thus tablets can be prepared by more than a single compression.

a) Multilayer or layered tablet :-

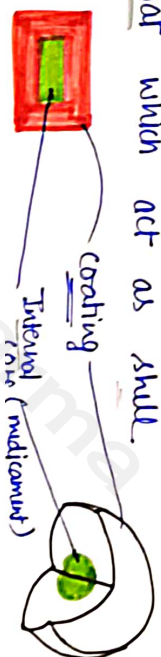
contains either two or more layered (components).



b) Compressed coated tablet :-

contains two parts

internal core (small tablet) and surrounding coat which act as shell.



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iii) Sustained tablet →

- These tablets are prepared for the purpose of long duration of action.
- So these tablets release their active ingredient over prolonged period.
- Prepared by hard compression.

iv) Enteric coated tablet (ECT) →

Those tablets which are designed to pass unchanged through the stomach and absorb into intestine.

- These tablets are prepared because drug substance get deactivate/destroyed by gastric acid or may cause irritation in stomach.
- It has delayed release properties.



v) Sugar coated tablet →

- Those tablets which has sugar coating by concentrated sugar solution.
- This coating are basically used to mask the tasting or smelling drugs.
 - Also protect drug and enhance appearance.

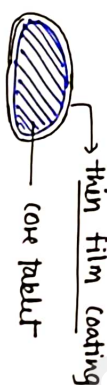


vi) Film coated tablet →

They are compressed tablets coated with a thin layer of polymer, forming a skin like film over the tablet.

eg Antibiotics etc -

- polymers → Hydroxypropyl cellulose etc -



vii) Chewable tablet →

Those tablets which are intended to be chewed by patients. easily administered

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to children and elder persons who may have difficulty in swallowing a tablet.

eg Antacid

- Basically larger in size



② Tablet used in Oral cavity :-

These tablets which are administered into mouth (orally) but not swallowing instead used in oral cavity.

i) Buccal tablets →

These tablets which are placed in

buccal pouch (between gums and lips) where they dissolved or disintegrate slowly and absorbed into blood without passing GIT.

eg. progesterone tablet

ii) Sublingual tablets →

These tablets are placed under the tongue where they dissolved or disintegrate slowly and absorb into blood without

passing GIT. eg. GTN (Glyceryl Trinitrate)

iii) Lozenges and Troches →

- These tablets which are commonly used to treat sore throat or to control coughing in common cold.

- These tablets are intended to be slowly dissolved in the oral cavity.

- Usually prepared by compression method or by candy moulding process.

eg. Sipsals, vices etc.

iv) Dental cone →

These are small tablet which insert in empty socket of teeth after extraction, prevent from bacteria

- These tablet contain an excipient like lactose, sodium bicarbonate and sodium chloride.
- Generally get dissolved 40-45 minutes time.

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③ Tablet administered by other routes :-

Those tablets which administered other than oral routes.

i) Implantation tablets →

• Those tablets which are placed under the skin or inserted by minor surgery.

• These tablets have slow absorption and provide continuous release of drugs for the months or a year.

• These tablets must be sterile implants

• made by heavy compression or fusion

• Uses → used for administration of hormones such as testosterone steroids for contraception.

eg

ii) Vaginal tablets →

Those tablets which inserted in vaginal cavity and dissolve slowly.

• these tablets are used to release antibacterial agents, antiseptics etc- to treat vaginal infections.

• these tablets are uncoated, bullet shaped, or oval tablets for ease of insertion.

④ Tablet used to prepare solution :-

Those tablets which firstly used to prepare solution, then this solution used for action.

i) Effervescent tablet →

Those tablet which dissolved in water and release CO_2 gas.

• These tablets are prepared by compressing granular effervescent salts such as sodium bicarbonate and citric/tartaric acid.

• Release of CO_2 in water produce disintegration of tablet & produce (increase) palatability and also give rapid action. eg. Disprin (Aspirin).

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ii) Dispensing tablet →

- These tablets are intended to be added to a given volume of water to produce a solution of given drug concentration.
- This is because tablet contains large amount of highly potent APIs required water to make solution.

eg Naproxen sodium etc.

iii) Hypodermic tablet →

- These tablets which are composed of drugs and water soluble ingredients.

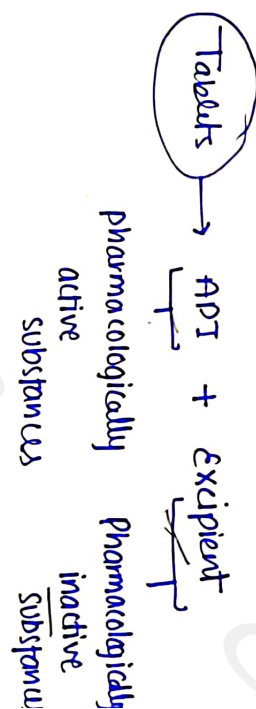
• These tablets are dissolved in sterile water or water for injection and administered by parenteral routes.

eg morphine sulphate etc.

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Excipient



- An excipient are those substance which has no therapeutic value, but are added with drug in formulation for—
 - help in production
 - improve patient acceptability
 - improve stability.
 - etc.

• following excipients are used in tablet formulation :—

- i) Diluent / filler
- ii) Granulating agent
- iii) Binding agent / Binder (Adhesive)

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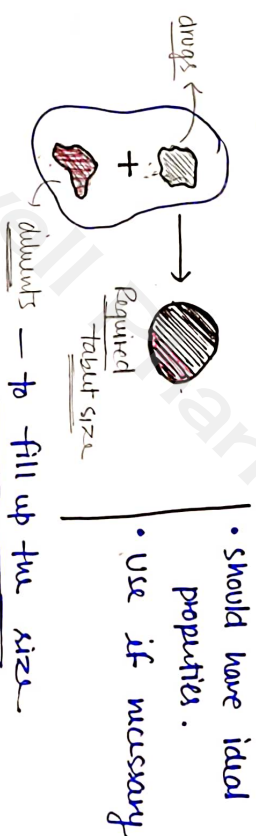
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- iv) Disintegrating agents
- v) Lubricants and Glidants
- vi) Adsorbing agents
- vii) Colouring, flavouring and sweetening agents.
- Ideal properties of excipients :-
 - Must inert in nature (physiologically inert).
 - Non-toxic & acceptable
 - Compatible with API.
 - low cost (not expensive).
 - Stable.
 - free from microbes.
 - color compatible
 - also do not alter the bioavailability of drugs.

i) Diluents →

Also known as filler or bulking agents.

- These are designed to make up the required bulk of the tablet when the drug dosage itself is inadequate (insufficient) to produce this bulk.



eg) Dextrose, starch, Sorbitol, lactose, Calcium phosphates, Mannitol, hydroxyl propyl methyl cellulose (HPMC).

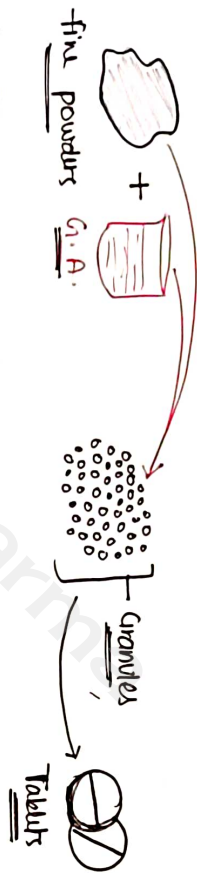
ii) Granulating Agents →

These are those substances which are used to convert the fine powders into granules.

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- These agents provided proper moisture to convert fine powder into damp mass which after passing through a sieve of suitable number forms granules.



eg. Tragacanth, mucilage of starch, fucacia, water alcohol etc—

iii) Binding Agents / Binders / Adhesive →

- Those substances or material which are added to a drug-filler mixture to ensure that excipients tightly bind with medicaments.
(to improve cohesive properties.)
- These agents are present in both dry or wet form. to form granules.

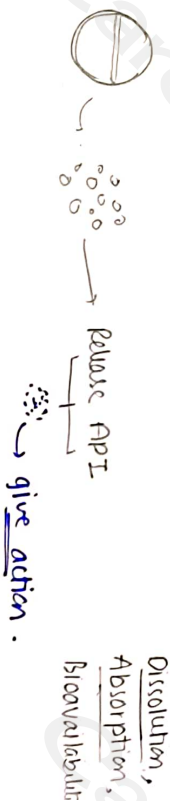


eg. Tragacanth & fucacia, Gelatin, Sucrose,

starch paste, cellulose etc—

iv) Disintegrating agents →

- These are those substances/materials which are added to a tablet formulation to facilitate its breaking/disintegration after administration in the G.I.T.



eg.

- Benzonite, Starch USP, Modified
- Starch (Primogel / Explotab) — 1-8%.
- Clay (veegum & Bentolite) — 10%.
- Used in parts (half use before completion tablets)

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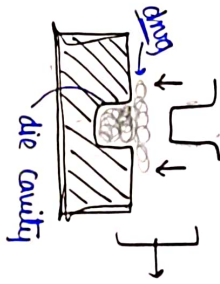
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v) lubricant & Glidants →

• lubricant → Reduce friction during tablet ejection between the walls of the tablet and the walls of the die cavity in which tablet formed.



— prevent sticking
eg stearic acid,
magnesium stearate,
Talc, surfactants—

• Glidants → Improve flow properties of granules & tablets—

eg Corn starch (5-10%), Talc (5%) & some silica derivatives.

* Both are added in tablet formulation to increase flow properties and improve the appearance of tablets and prevent sticking

vi) Absorbing agents →

• Those substance/material which are used (added) in tablet formulation to absorb the moisture.

• Also present in the form of small packets—

eg. Silica gel etc—

vii) • Colouring agents → These agents are added to make tablet more aesthetic (good) in appearance.

• used to helps in identification of tablets.

eg Dyes & lakes — D & C Red 22, FD & C Yellow 5.

• Flavouring agents → These agents are added to make the unpleasant taste

of drugs.

— increase acceptability of tablets.

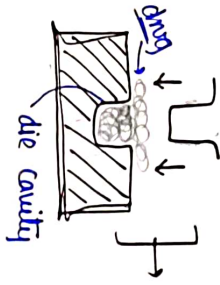
eg mint, orange — vicks, ginger mouthwash etc—

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eg Dyes & lakes - D & C Red 22, FD & C Yellow 5.

• flavouring agents → These agents are added to mask the unpleasant taste

of drugs.

- increase acceptability of tablets.

eg mint, orange — vicks, ginger mouthwashes etc -

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• Sweetening agents →

Added in formulation to mask the bitter taste. → ↑ acceptability
eg Sucrose, lactose, mannitol, saccharin etc.

viii) Preservatives →

Those substance which are used ~~to~~ in formulation to prevent drug from microbes and increase its stability.
eg Methyl paraben, ethyl paraben etc..

FORMULATION OF TABLETS

Also known as tablet manufacturing process.

- Through which we prepare tablets
- There are three methods for this:—

— Granulation
 ① wet granulation ③ Direct
 ② Dry granulation Compression.

• Granulation →

It is a process in which fine powders of drugs is converted into small granules, which further used to prepare tablets through compression.

① wet granulation →

In this process, we use binders solution in wet form.

- The steps required for this:—

- i) Mixing of API & excipients
- ii) Preparing a damp mass using binder solⁿ
- iii) Screening — granulation
- iv) Drying
- v) Sizing the granulation by dry screening
- vi) Lubrication
- vii) Compression

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i) Mixing →

• firstly we have to weigh the required amount of API (medicament) and all excipients

• Now, these excipients such as diluent, and half amount of disintegrating agents are mixed with API through blender (V-cone blender).

ii) Preparing a damp mass →

• Now, in this mixture we have to add a solution of binder (in wet-form) to make a damp mass.

iii) Wet screening →

Now, this damp mass is screened using sieve with the help of hand or suitable equipment, and forms granules.

iv) Drying :-

Now these wet granules are dried in hot air oven at 60°C (adjust).

v) Dry screening :-

After drying these granules are screened through a sieve to get uniform size granules.

vi) Lubrication →

Now these granules mixed with glidant then add lubricant in it and also add remaining disintegrants.

vii) Compression :-

Now compress this granules into desired punching machine to give proper shape and size to the tablets.

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• Advantages -

- cohesives and compressible properties are improved.

- Good flow properties.

- Better uniformity and good distribution.

• Disadvantages -

- Costly

- more steps required more labour + time

②

Dry granulation

[Slugging/double compression method].

In this process, we use binders in dry form.

• The steps required for this :-

i) Mixing

ii) Slugging (first compression)

iii) Crushing (breaking)

iv) Screening

v) mixing (lubricant, disintegrating) vi) compression.

i) Mixing →

• first weigh the desired (required) amount of API and all excipients.

• Now add these excipients with API and also add half amount of lubricant with dry binder. (in binder) ^(to prevent sticking & ↑ flow)

ii) Slugging →

Now these mixture are compressed into large flat tablets (pellets) known as "slug". (about 1 inch in diameter).

iii) Crushing →

Now these slugs again breaks into small particles using mill (suitable).

iv) Screening →

Now these small particles screened to produce uniform sized granules.

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v) Mixing →

Now, Add remaining amount of lubricant.

and also add glidant and disintegrating agents to granules and mixed it into blenders.

vi) Compression →

Now compress these granules into desired punching machine to give proper shape and size to the tablets and produce tablets.

* Compression also depend on hardness of tablets.

• Advantages →

- Suitable for moisture and heat sensitive material.
- Required less equipment & minimum floor space.
- Better disintegration.

• Disadvantages →

- create more dust than wet granulation
- Not permit uniform color distribution than wet granulation.
- required heavy duty tablet press to form slug

③ Direct compression method →

In this process, drug powder

material is directly compressed into tablets

• This process is suitable for those drug substance which has good compressible and flow properties.

• Now, firstly weigh the API (medicament) and some suitable excipients such as Diluents (eg spray-dried lactose, crystalline maltose etc.), Disintegrants (eg starch, sodium carboxymethylcellulose), lubricant (eg magnesium stearate) etc.

• Then mixed and prepare tablets through compression.

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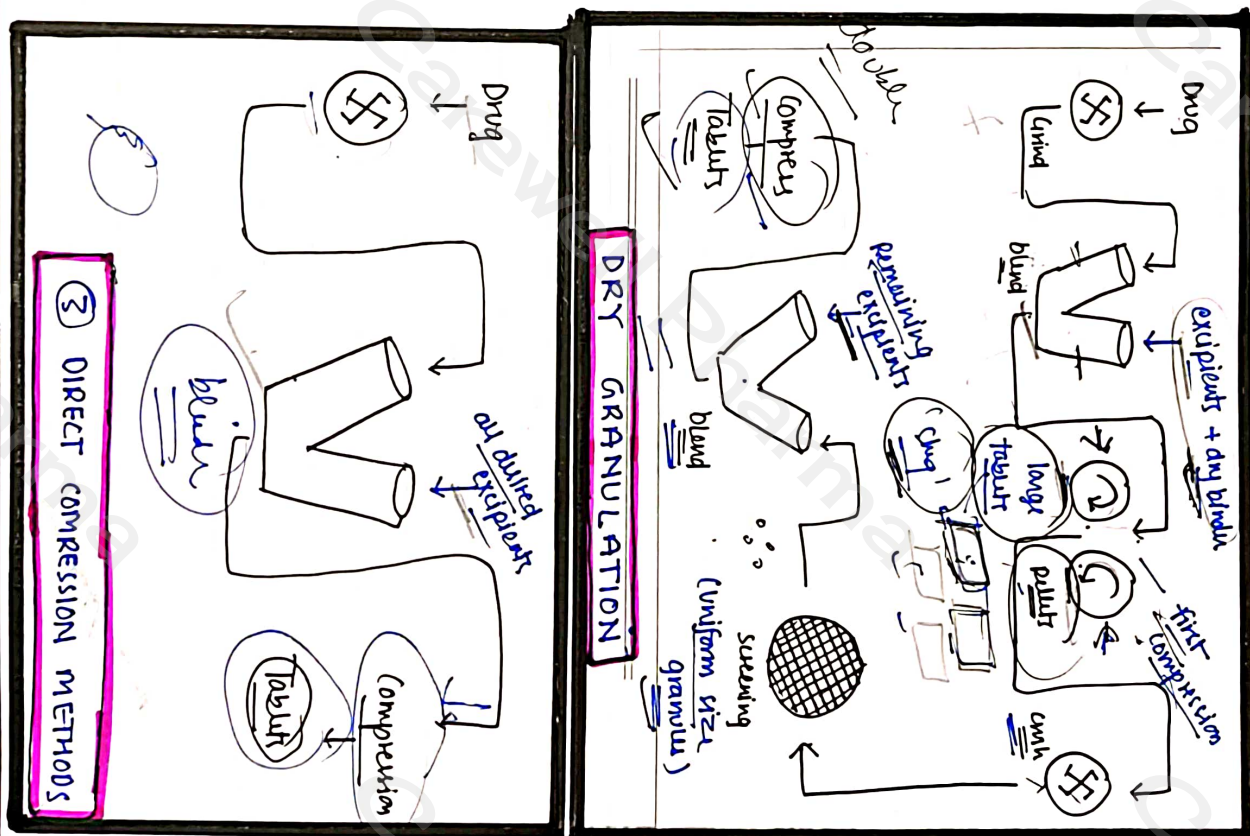
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- Advantage →
 - simplest & require less time & equipment
 - cheap
 - faster dissolution rate due to high disintegration (no involvement of moisture).
- Disadvantage →
 - Not suitable for high dose drugs (not easily compress).
 - Sometime not blended uniform.
 - Not suitable for all drugs.

Rough Easy
Diagram for these
methods :-

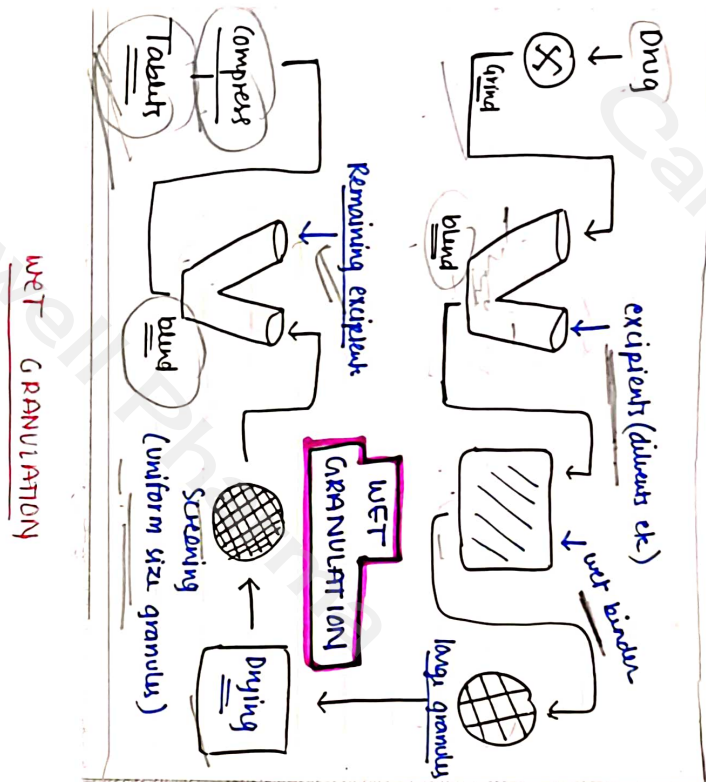


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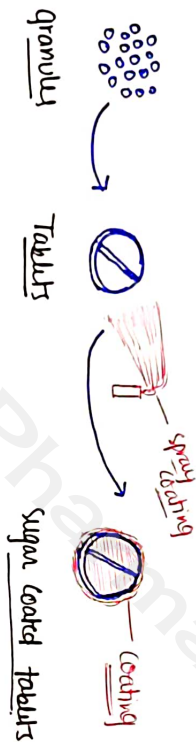
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TABLET COATING

It is a process in which the tablet surface is covered by layer to improve the appearance and to mask the unpleasant taste and odour.



• Causes/Reason for tablet coating →

- To mask unpleasant taste and odour.
- To provide physical/chemical protection to drugs.
- To improve the appearance of tablets.
- To produce the sustained release product.
- To produce enteric coated tablets.

• Types of coating →

mainly there are three types of

coating:-

- i) Sugar coating
- ii) film coating
- iii) enteric coating

i) Sugar coating - This type of coating is

used to mask the unpleasant

taste of tablets and enhance acceptability of tablet.

- deposit sugar layer on the surface of tablet

- The various steps involved in sugar coating

- i) Sealing (water proofing)
- ii) Subcoating
- iii) Smoothing
- iv) coloring
- v) Polishing.

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i) Sealing (water proofing) →

- It is a seal coating over the tablets which prevents tablets from water penetration or moisture.
- increase stability + safety of drugs.

eg Shallac, Zein (a corn protein derivative) etc. are used as sealants polymer for this

ii) Subcoating →

- It is performed to round the shape of the tablet edges.
- This step increases the weight of the tablets
- material used for subcoating
- Alacia, Gelatin - (levination process)
- gum/sucrose solution - (suspension process)
- Also apply dusting powder to prevent sticking.
eg Calcium carbonate, titanium dioxide, nonwater soluble etc.

iii) Smoothing (syruping) →

- This step is applied to smooth out subcoated rough surface and increase tablet bulk to desired size.

- Several coat of simple syrup solution (60-70%) are applied.

- Now, this syrup contain sugar and acacia, gelatin, pigments, starch etc.

iv) Coloring →

- It gives the color and finished size to the tablets
- Sugar coating solution consists of 70% syrup and other coloring pigments.
- water soluble dyes or water insoluble pigments may be used.

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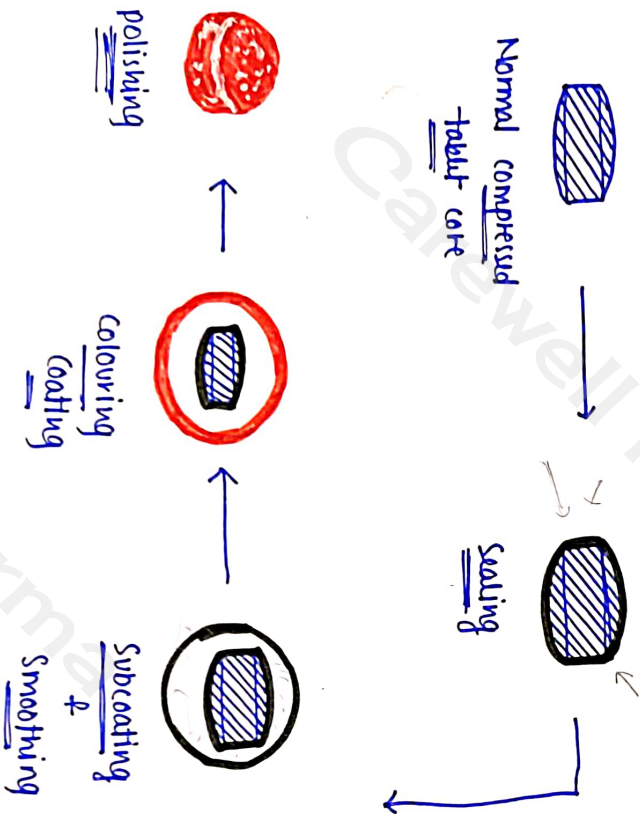
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v) Polishing → (in standard coating pan)

- Generally sugar-coated tablets are dull in appearance, so polishing give the surface shine and tablet elegance.
- eg busuax, caranava and candilia wax are used for polishing.

SUGAR COATING



ii) film coating →

It is a process of covering of tablets with a thin polymer layer.

- the thickness of this coating is usually about 40-100 μm.

- Advantages →

- protects drugs from atmospheric effects
 - less time consuming because it is a single stage process
 - not so much affect tablet disintegration time
 - not so much increases the tablet's weight.
- material used in film coating →

- Polymer (film former)
- Solvents
- Plasticizers
- Colourants
- miscellaneous agents.

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- Polymer (film-former) → These are used to improve product appearance and to protect the drug substance from atmospheric conditions.
eg Hydropropyl methyl cellulose (HPMC),
Methyl hydroxyethyl cellulose, Povidone etc--
 - Solvents → used to dissolve the polymer.
eg water, methanol, ethanol, acetone etc..
 - Plasticizers → These are added to film-coating formula to improve the flexibility and processability polymer-film.
eg Polyethylene glycol (PEG), glycerol, phthalic ester,
castor oil, triclin etc--
 - Colorants → These are added to film coating formula (opacifier) to improve product appearance.
eg Sunset yellow, tartaric acid, erythrosine and some other insoluble pigments.
 - Miscellaneous agents → eg flavours & sweeteners,
surfactants, antioxidants, antimicrobials etc....
- Ideal properties of these materials →
 - It should be soluble in the solvent of choice for coating preparation.
 - It should have the solubility required for intended use
 - Able to produce elegant product
 - stable in presence of heat, light, moisture etc..
 - compatible, inert, non-toxic.
 - form thin flexible film.
 - iii) Enteric coating →
Basically, it is one type of film coating
It is a process of covering of tablets with a thin enteric polymer layer which protect the tablet core from disintegration in the acidic environment of the stomach.
 - These tablets are dissolved in intestine.

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- material used
- Enteric-coating polymers → these are pH sensitive polymers.
 - insoluble at acidic pH (stomach)
 - soluble at basic pH (basic small intestine).
- [eg] Hydroxypropyl methylcellulose phthalate, Polyvinylacetate phthalate (PVAP), Acrylate polymers (Eudragit L & Eudragit S) etc...
- Reason for this —
 - to protect "acid labile drug" (those drugs which easily destroyed in acidic environment). eg Enzymes & Antibiotics.
 - If drugs cause irritation to the stomach.
 - Drugs intended for local action in intestinal eg intestinal antiseptics.
 - to provide a delayed-release components for repeat-action tablets.



→ coating material / composition.

- Coating materials → These are those materials which are used for coating or applied on the surface of tablets.
 - eg Sugar, Shallac, polymers such as Hydropropyl methyl cellulose, wax coating. etc....
- formulation of coating composition → These are completed all material which are used for coating.
 - eg film coating (non-enteric)
 - Polymers, solvents, plasticizers, colourants etc...
- (all are defined in film coating).
- Methods of coating → These are techniques which are used for coating:—
 - these are followings:—
 - Sugar coating
 - film coating } already discussed

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- Dip coating → This type of coating is applied by dipping core tablet in coating liquids and then dried.
 - Brush coating → In this type, coating is applied by brushing (required experienced person)
 - Spraying → Versatile, economical, fast coating process for different type, sized, shaped parts including large surface areas.
 - Use different types of spray gun for this.
 - Coating is applied by spraying.
 - Electrostatic coating → In this type, a substrate is present at which strong electrostatic charge is applied while coating material containing opposite charge is sprayed over it to get uniform coating.
 - Compression coating → This is generally a tablet within a tablet, in which coating material is applied on the tablet core by compression using tablet machine.
 - generally used for make the taste and to delayed the release of drug.
 - Vacuum film coating → In this type, tablet are placed in hot ~~air~~ ^{water} jacketed pan, now vacuum is applied by scaling, then coating solⁿ is applied by hydraulic spray system.
 - & heated pan remove moisture through evaporation.
-
- ✓ Sugar coating
 - ✓ film coating
 - ✓ Dip coating
 - Brush coating
 - ✓ Spray coating
 - Electrostatic coating
 - ✓ compression coating
 - Vacuum film coating

Methods of
coating

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- Equipment employed in coating →

(coating equipments are divided into three types:—

- i) Standard coating pan (conventional)
- ii) Perforated coating pan
- iii) fluidized bed coater (air suspension)

- i) Standard coating pan →

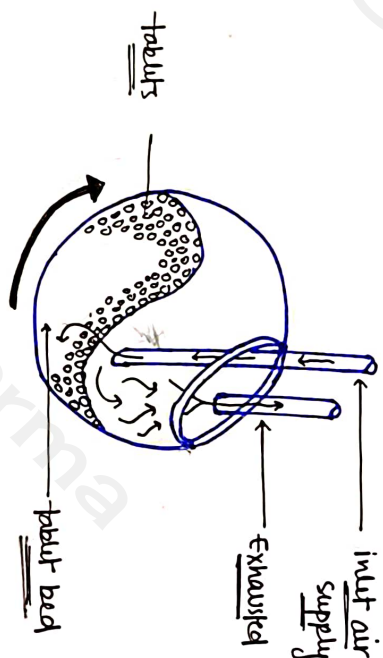
- It is also known as

Conventional coating pan (pan coater).

- It is most popular because it is easy to operate and drug can be easily removed due to wide mouth of pan.

- Construction & Working :-

- It consists of a circular pan made of metal mounted on a stand.
- The pan is slightly tilted to an angle of about 45° to the bench top.



- The pan is about 8-60 inches in diameter.
- Now, tablets are introduced onto the tablet bed surface.
- Then, coating solution are applied to the tablets by spraying the material on to the rotating tablet bed.
- Also, The heated air is directed into the tablet bed by inlet air supply.
- air leaves by exhausted duct.
- The temperature of hot air should be maintained.

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It is of three types -

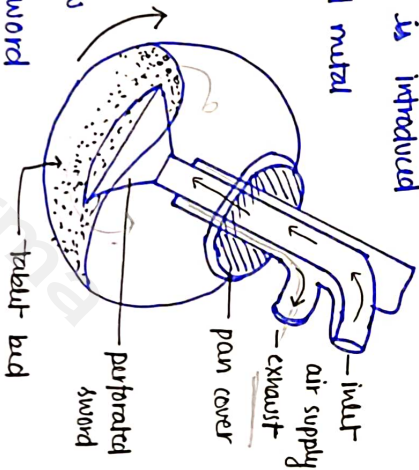
- i) Pellegriani pan system (baffled pan & diffuser)
- ii) Immersion sword system
- iii) Immersion tube system

i) Pellegriani Pan - Also known as Baffled Pan and diffuser.

- In this, drying air is uniformly distributes over the tablet bed surface & improves drying efficiency of standard coating pan.

ii) Immersion sword system -

- In this, drying air is introduced through a perforated metal sword device which immersed in the tablet bed and the drying air flow upward from the sword through the tablet bed.

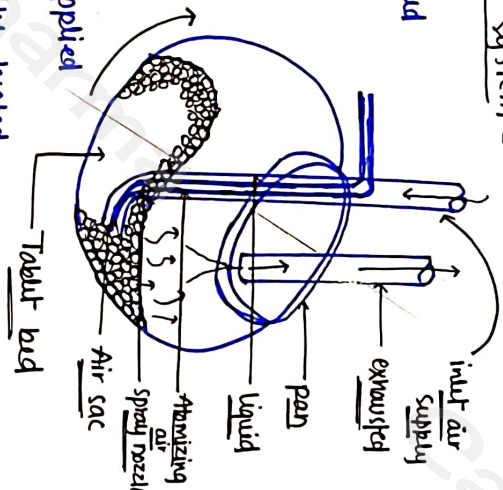


iii) Immersion tube system -

- The tube is immersed in the tablet bed
- Also, a spray nozzle is built in the tip of the tube
- Now, in this the coating solution is applied simultaneously with the heated air from the immersed tube
- The drying air flows upward through the tablet bed and is exhausted by a conventional duct.

2) Perforated pan system →

- It consist of a perforated or partially perforated drum that is rotated on its horizontal axis in a closed chamber.
- The coating drum conduct various spray nozzles that atomize the coating soln.



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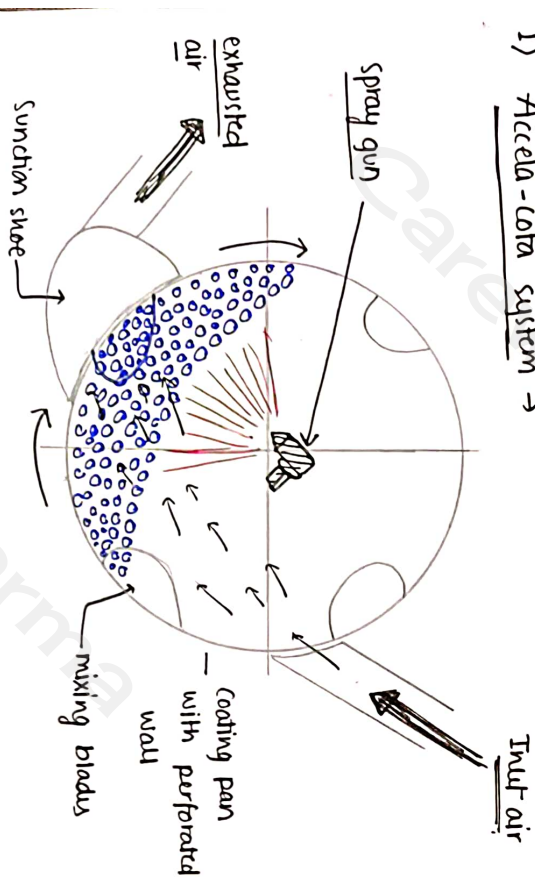
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- Drying air is passed through the tablet and is exhausted through perforations in the drum.
- Efficient drying system with high coating capacity (useful for both sugar + film coating).

— It is of four types :-

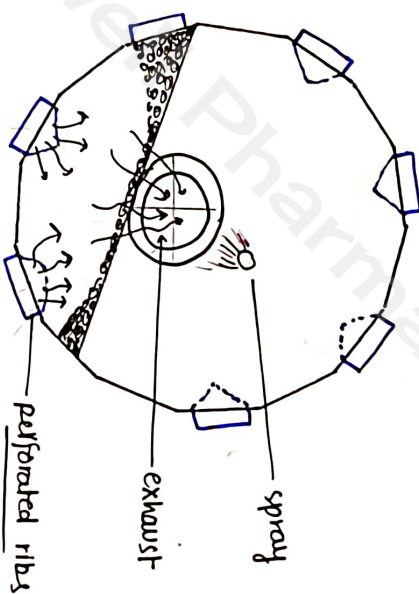
- 1) Accela-cota system
- 2) Dria coater pan
- 3) Hi-coater system (pan)
- 4) Clatt coater

1) Accela-cota system →



- Drying air is directed into the drum, is then passed through the tablet bed, and then is exhausted through perforation in the drum.

2) Dria coater pan system →



- In this coater,

drying air is introduced through the hollow perforated ribs located on the inside periphery of the drum.

- When drum rotates, the spray nozzle atomizes coating solution to the tablets from the top section.
- Air is exhausted from the back of the pan.

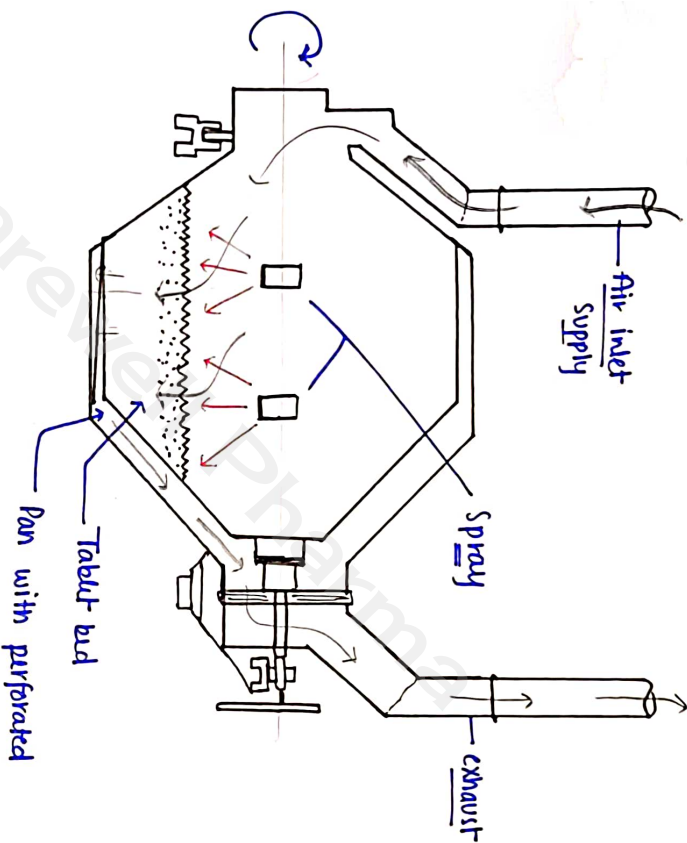
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3) Hi-coater system →



- The drying air is passed through tablet bed and then exhausted through the perforation below the coating drum.

4) Glatt coater → Drying air can be directed from inside the drum through the tablet bed and out an exhaust duct.

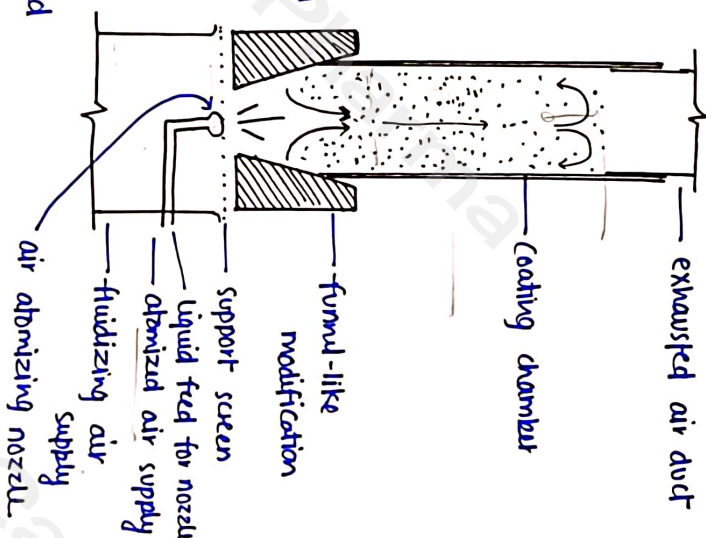
iii) fluidized bed coater (air suspension) →

- It consists of a vertical cylinder.
- It is also highly efficient coating system.

- The air enters the centre of the column causing tablets rise in the centre

- Now, coating solutions are applied from spray nozzle which is located at bottom or top of chambers into a fluidized bed.

- Now there are two types of spray systems:—
 < high pressure airless
 low pressure air atomized.



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- High pressure airless →
 - liquid is pumped at high pressure (250 - 3000 psig) through a small orifice (0.009 inch - 0.020 inch) in fluid nozzle which results in a finely divided spray.
- Low pressure air atomized →
 - liquid is pumped through a somewhat larger orifice (0.020 inch - 0.60 inch) at relatively low pressure (5 - 50 psig)
 - low pressure air contact the liquid stream (10 - 100 psig) at the tip of the atomizer and finely divided spray is produced.

- Defects in tablet coating →
 These are the problem occurs during and after coating the tablets.

- 1) Chipping
- 2) filling
- 3) bridging
- 4) Roughness
- 5) Blooming
- 6) Sticking and picking
- 7) Color variation
- 8) Glistering →

The high temperature occurs during process cause rapid evaporation of solvent and produce an un-smooth coated surface with number of uneven spots are called blister.

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• treatment → milder drying condition are suggested in this case

2) chipping →

It occurs when the film becomes dented and chipped from the edges of the tablet.

• Cause → decrease in rotation speed + incorrect amount of plasticizer during coating

• Treatment → overcome by adjusting the proportion of plasticizer.

3) filling →

It is caused by applying too much solution, resulting in thick film that fills and narrows the bisect. (fill monogram etc.)

• Treatment → careful monitoring of the fluid application rate and thorough mixing of the tablet in the pan prevent filling.

4) bridging →

During drying the film may shrink and pull away from the sharp corners of an infatigation or bisect resulting in a bridging of the surface depression

• Treatment - problem with the formulation (maintain it).

5) Roughness →

A rough or gritty surface is observed when the coating is applied by spray. Some droplets may dry too rapidly before reaching the tablet bed cause deposition of spray dried particles.

• treatment - moving the nozzle closer to the tablet bed or reducing the degree of atomization.

6) Blooming →

tablet become dull or hazed after prolonged period of storage at high temp.

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- It is due to using too much plasticizer.
- Treatment → Reduced by decreasing the concentration of the plasticizer.

7) Sticking and Picking →

- Over wetting and excessive film thickness causes tablets to stick to each other or to the coating pan.
- part of the film stick to the pan + giving a picked appearance to the tablet.
- treatment → A reduction in the liquid application rate or increase in the drying air temp and air volume.

8) Color Variation →

- Problem caused by processing condition or the formulation can cause color variation.
- treatment - overcome by geometry mixing + using mild dry conditions.

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Quality Control tests - Tablet Evaluation Parameters

These tests are done to evaluate the each steps of formulation and also finished products to obtain a good quality of the products.

- there are several test for evaluation of tablets:—

- | | | |
|------------------------------|---|---------------------------|
| ① General appearance | } | <u>Non-official tests</u> |
| ② Size & shape | | |
| ③ Thickness test | | |
| ④ Hardness test | | |
| ⑤ friability test | | |
| ⑥ Weight variation | } | <u>Official tests</u> |
| ⑦ Content uniformity | | |
| ⑧ <u>Dissolution test</u> | | |
| ⑨ <u>Disintegration test</u> | | |

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General appearance →

The general appearance of a tablet is its identity and overall "elegance" which is essential for consumer acceptance.

- It involves - size, shape, colour, odour, taste, surface texture

② Size & shape →

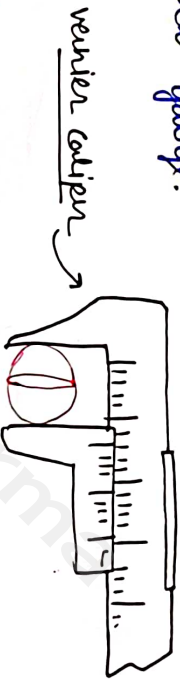
It is depends upon die and punches used for making tablets.

- Generally they are biconvex, concave or flat.

③ Thickness →

Tablet thickness should be controlled within $\pm 5\%$ variation of standard value.

- It is measured by vernier caliper or steel gauge.

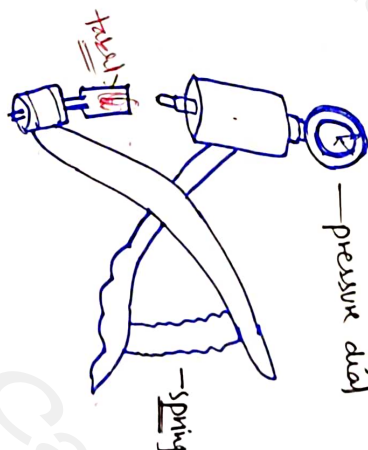
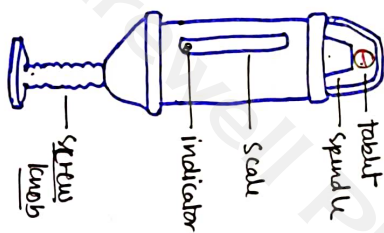


④ Hardness test →

Hardness generally measures tablet crushing strength (the force required to break a tablet).

- It is measured to determine the need for adjustment of pressure for compression.

— Hardness testers →



• Monsanto hardness tester

• Pfizer hardness tester

- too hard tablet - not disintegrate easily
- too soft tablet - cause problem during handling / packing.

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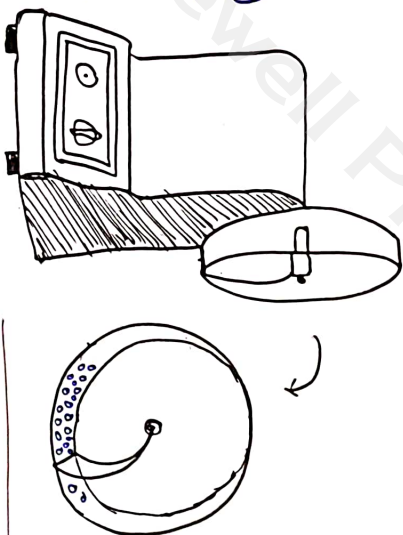
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frictability test →

- This test is official in ISD but not in IP.
- This test is applicable for uncoated compressed tablets.
- To check physical strength of tablets during handling and transportation.
- The instrument used is "Rodet's friabilator".

Procedure →

- Take 20 tablets randomly & weight (initially)
- Now drop tablet in apparatus at distance of 6mm
- Speed - 25 RPM
- total → 4 min, 100 rounds
- Now, again weight the tablets.
- Conventional tablets → Variation is not more than 0.5 - 1%.



$$\% \text{ loss} = \frac{W_1 - W_2}{W_1} \times 100$$

$W_1 = \text{initial weight}$
 $W_2 = \text{final weight}$

⑥ weight variation (uniformity of weight)

- The weight of tablets measured routinely is help to ensure that tablet contains the proper amount of drug
- Take 20 tablets and weighed individually.
 - calculate average weight and compare the individual tablet weight to the average.
 - Not more than 2 tablets are outside the percentage limits.

Average weight of tablet		% deviation
fit per USP	fit per IP	
130mg or less	80mg or less	10%
130mg - 324mg	80mg - 250mg	7.5%
more than 324	more than 250	5%

⑦ Content uniformity →

To determine the active ingredient by the method in assay

— procedure —

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30 tablets are randomly selected in which atleast 10 of them are assayed individually.



9 of the tablets must not contain less than 85% or more than 115% of the labelled drug content.

10th tablet may not contain less than 75% or more than 125% of the labelled content.



If these conditions are not met, then the tablet remaining from the 30 must be assayed individually and



Non may fall outside of the 85% to 115% range.

⑧ Dissolution test →

It is an official test.

• Dissolution → It is a process by which a solute dissolve into a solvent or solution.

So, This test is done to predict the time required for a given amount of drug to be released into solution. [Drug released check]
Also helps to determine the bioavailability of drug.



• Apparatus →

It consist of a cylindrical vessels with hemispherical bottom and made up of transparent glass having 1000 ml capacity. The vessel is fitted with a cover having 4 holes, one for

Stirrer, one for thermometer and rest two for sample

Type I → Paddle type
Type II → Basket type } Acc. to I.P.

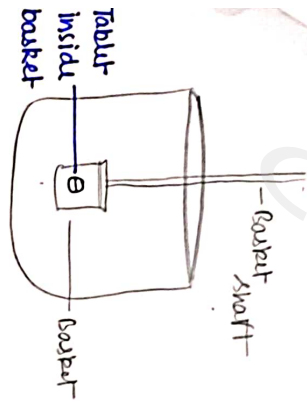
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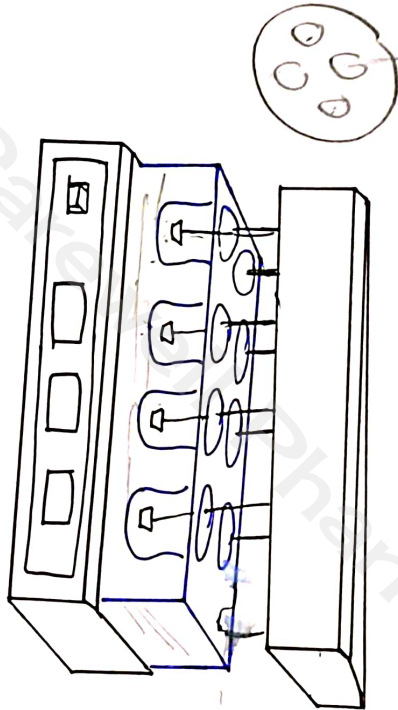
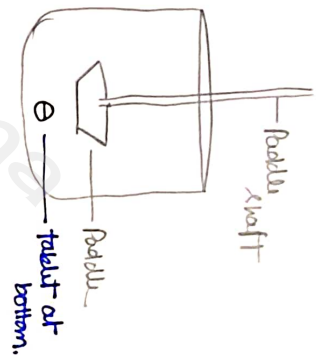
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basket type



Paddle type



DISSOLUTION TEST APPARATUS

• Procedure →

- Place 1000 ml of dissolution medium (dist. water, 0.1 N HCl, Phosphate buffer) into vessel
- Turn this vessel immersed in water bath and maintain temp. $37^{\circ}\text{C} \pm 0.5^{\circ}\text{C}$.

- Now placed tablet(s) acc. to type [basket/paddle].

- Starts and speed maintain at 100rpm acc to monograph (in I.P.)

- After specified time interval (acc to monograph in I.P.) take 5-10 ml of sample (or specified) from vessel and refill vessels with 5-10ml fresh medium.

- Now, this sample is tested by U.V. spectroscopy or some other suitable analytical method.

- Now, find the drug amount which released in sample and match with standards (I.P.)

- If all 6 tablets passed test then okay, but ^{8x5%} if one/two fails then do this again for additional 6 tablets → $\geq Q$ & $Q-15\%$

- If tablets still fails the test, then an additional 12 tablets are tested.

• Now, the tablets are acceptable if the average of all 24 tablets are $\geq Q$ & not more than 2 tablets are less than $Q-15\%$.
 monograph tolerance repeat

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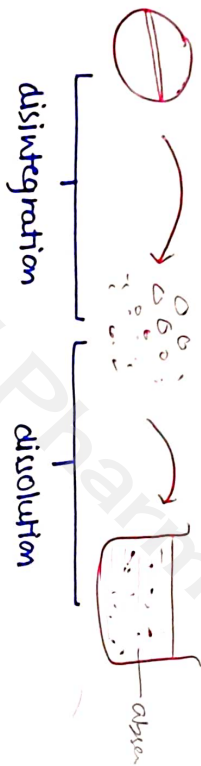
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Disintegration test →

- Disintegration → It means to break the tablet into smaller particles after swallowing.
- The time required to disintegrate the tablet is called disintegration time.



- Apparatus → Disintegration test apparatus.
- It consists of a water bath which is filled upto mark mentioned.
- A basket rack holding six plastic tubes open at top and the bottom is covered with a 10 mesh screen.
- Procedure →
- Now, The basket rack assembly is suspended in liquid medium in 1000 ml beaker and

place this beaker into water bath and maintain the temp. of liquid at 37°C.

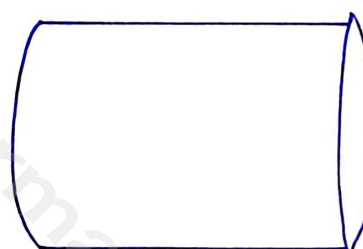
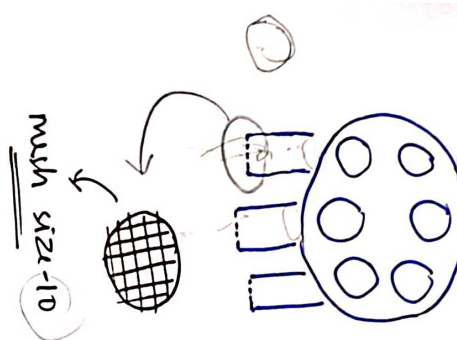
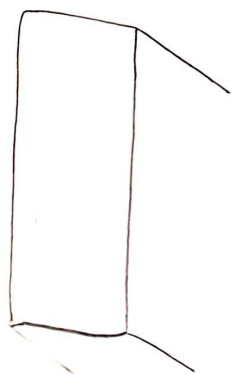
- Now, place one tablet into each tube.
- The assembly moves up and down at specified rate (30 RPM).
- A cylindrical disk made of transparent plastic is also placed over the tablet.
- The disc also generate little pressure on tablet.
- Now, noted down the disintegration time.
- Uncoated Tablet → Disintegrate within 15 minutes.
- Sugar coated tablets → " within 60 minutes.
- film coated tablet → " within 30 minutes.
- Enteric coated tablet → " within 60-120 minute

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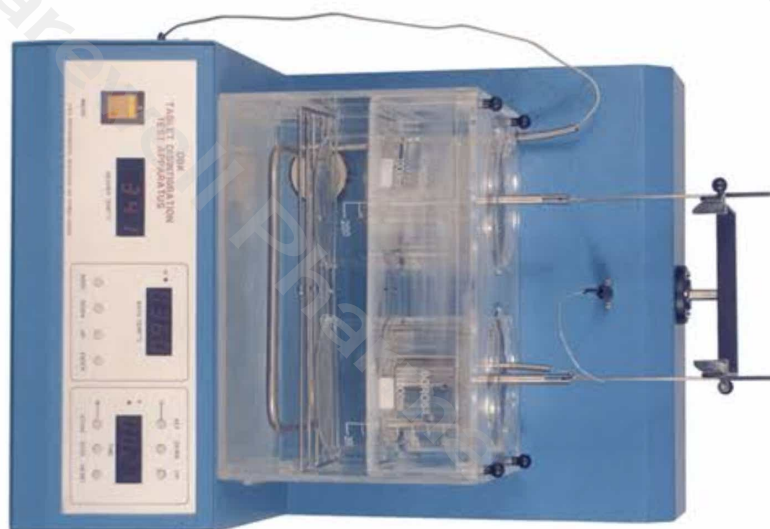
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— 1000 ml
beaker

Disintegration Apparatus



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